

Program 11: Creating a Ansible WebApp with the help of Tomcat Server

Step 1: Run the command in order to create a project with the name “**MyMavenAnsibleWebApp01**”

mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenAnsibleWebApp01 -DarchetypeArtifactId=maven-archetype-webapp -DinteractiveMode=false

```
sumanth@sumanth-VirtualBox:~/devops$ mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenAnsibleWebApp01 -DarchetypeArtifactId=maven-archetype-webapp -DinteractiveMode=false
[INFO] Scanning for projects...
[INFO] ----- org.apache.maven:standalone-pom -----
[INFO] Building Maven Stub Project (No POM) 1
[INFO] ----- [ pom ] -----
[INFO]
[INFO] >>> mvn-archetype-plugin:3.4.1:generate (default-cli) > generate-sources @ standalone-pom >>>
[INFO]
[INFO] <<< mvn-archetype-plugin:3.4.1:generate (default-cli) < generate-sources @ standalone-pom <<<
[INFO]
[INFO] --- mvn-archetype-plugin:3.4.1:generate (default-cli) @ standalone-pom ---
[INFO] Generating project in Batch mode
[INFO] -----
[INFO] Using following parameters for creating project from Old (1.x) Archetype: maven-archetype-webapp:1.0
[INFO] -----
[INFO] Parameter: basedir, Value: /home/sumanth/devops
[INFO] Parameter: package, Value: com.example
[INFO] Parameter: groupId, Value: com.example
[INFO] Parameter: artifactId, Value: MyMavenAnsibleWebApp01
[INFO] Parameter: packageName, Value: com.example
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] project created from Old (1.x) Archetype in dir: /home/sumanth/devops/MyMavenAnsibleWebApp01
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 1.640 s
[INFO] Finished at: 2026-05-09T23:31:02+05:30
[INFO] -----
```

Step 2: Update the **pom.xml** with the relevant dependencies and the relevant plugins

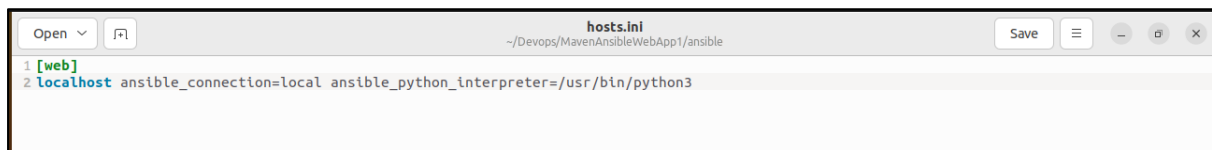
```
Open pom.xml Save
~/Devops/MyMavenAnsibleWebApp1
1 <project xmlns="http://maven.apache.org/POM/4.0.0" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xsi:schemaLocation="http://
  maven.apache.org/POM/4.0.0 http://maven.apache.org/maven-v4_0_0.xsd">
2   <modelVersion>4.0.0</modelVersion>
3   <groupId>com.example</groupId>
4   <artifactId>MavenAnsibleWebApp1</artifactId>
5   <packaging>war</packaging>
6   <version>1.0-SNAPSHOT</version>
7   <name>MavenAnsibleWebApp1</name>
8   <url>http://maven.apache.org</url>
9   <dependencies>
10    <dependency>
11      <groupId>junit</groupId>
12      <artifactId>junit</artifactId>
13      <version>3.8.1</version>
14      <scope>test</scope>
15    </dependency>
16    <dependency>
17      <groupId>javax.servlet</groupId>
18      <artifactId>javax.servlet-api</artifactId>
19      <version>4.0.1</version>
20      <scope>provided</scope>
21    </dependency>
22  </dependencies>
23  <build>
24    <plugins>
25      <plugin>
26        <groupId>org.apache.maven.plugins</groupId>
27        <artifactId>maven-war-plugin</artifactId>
28        <version>3.4.0</version>
29      </plugin>
30    </plugins>
31  </build>
32  <finalName>MavenAnsibleWebApp1</finalName>
33 </project>
```

Step 3: Run **mkdir ansible** to create an new directory and **cd ansible** to get into that directory

```
sumanth@sumanth-VirtualBox:~/devops$ mkdir ansible
sumanth@sumanth-VirtualBox:~/devops$ ls
ansible  MyMavenAnsibleWebApp01
```

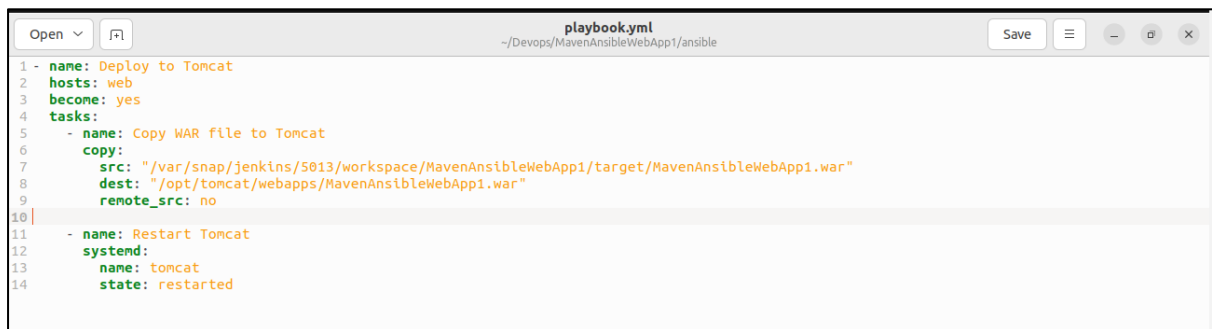
Step 4: Create two new files **hosts.ini** and **playbook.yml** and put the relevant code

```
sumanth@sumanth-VirtualBox:~/devops$ cd ansible
sumanth@sumanth-VirtualBox:~/devops/ansible$ gedit hosts.ini
sumanth@sumanth-VirtualBox:~/devops/ansible$ ls
hosts.ini
sumanth@sumanth-VirtualBox:~/devops/ansible$ gedit playbook.yml
sumanth@sumanth-VirtualBox:~/devops/ansible$ ls
hosts.ini  playbook.yml
sumanth@sumanth-VirtualBox:~/devops/ansible$
```



The screenshot shows a text editor window titled 'hosts.ini' with the following content:

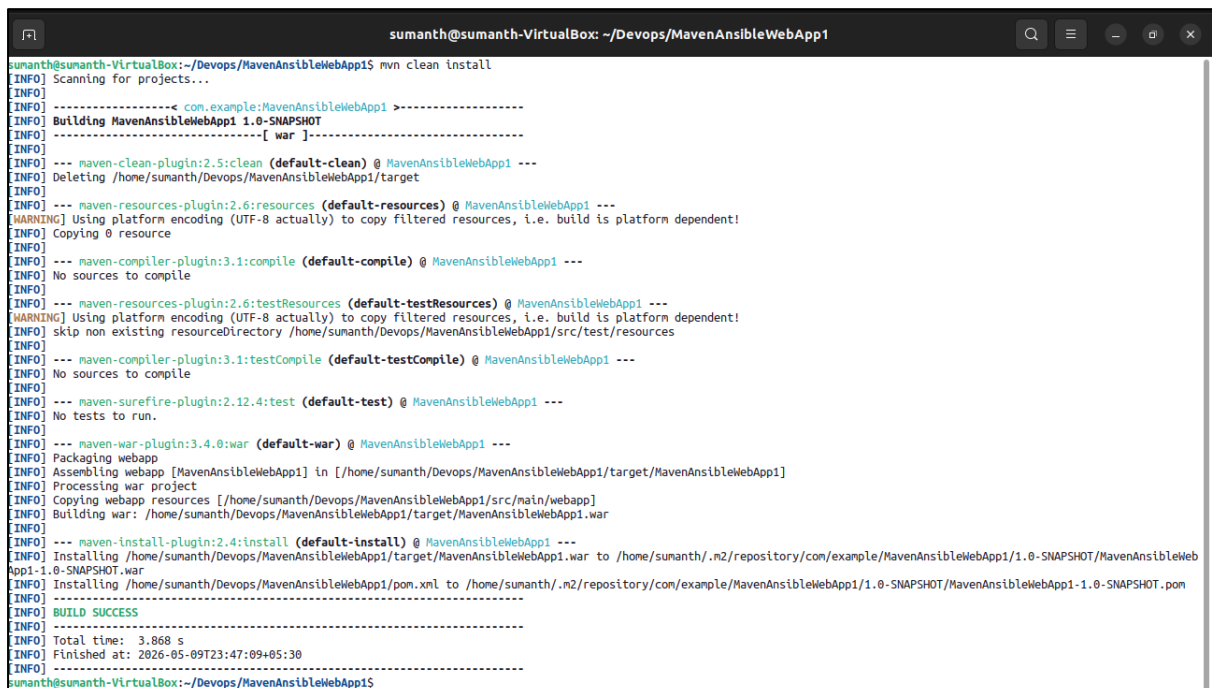
```
1 [web]
2 localhost ansible_connection=local ansible_python_interpreter=/usr/bin/python3
```



The screenshot shows a text editor window titled 'playbook.yml' with the following content:

```
1 - name: Deploy to Tomcat
2   hosts: web
3   become: yes
4   tasks:
5     - name: Copy WAR file to Tomcat
6       copy:
7         src: "/var/snap/jenkins/5013/workspace/MavenAnsibleWebApp1/target/MavenAnsibleWebApp1.war"
8         dest: "/opt/tomcat/webapps/MavenAnsibleWebApp1.war"
9         remote_src: no
10
11  - name: Restart Tomcat
12    systemd:
13      name: tomcat
14      state: restarted
```

Step 5: Run **mvn clean install** in order to compile and create a Jar file



The screenshot shows a terminal window with the output of the 'mvn clean install' command. The output includes various Maven lifecycle phases and their details, such as scanning for projects, building the webapp, copying resources, compiling, testing, and packaging the war file. The final output is 'BUILD SUCCESS'.

```
sumanth@sumanth-VirtualBox:~/Devops/MavenAnsibleWebApp1$ mvn clean install
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.example:MavenAnsibleWebApp1 >-----
[INFO] Building MavenAnsibleWebApp1 1.0-SNAPSHOT
[INFO] -----[ war ]-----
[INFO]
[INFO] --- maven-clean-plugin:2.5:clean (default-clean) @ MavenAnsibleWebApp1 ---
[INFO] Deleting /home/sumanth/Devops/MavenAnsibleWebApp1/target
[INFO]
[INFO] --- maven-resources-plugin:2.6:resources (default-resources) @ MavenAnsibleWebApp1 ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] Copying 0 resource
[INFO]
[INFO] --- maven-compiler-plugin:3.1:compile (default-compile) @ MavenAnsibleWebApp1 ---
[INFO] No sources to compile
[INFO]
[INFO] --- maven-resources-plugin:2.6:testResources (default-testResources) @ MavenAnsibleWebApp1 ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/sumanth/Devops/MavenAnsibleWebApp1/src/test/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.1:testCompile (default-testCompile) @ MavenAnsibleWebApp1 ---
[INFO] No sources to compile
[INFO]
[INFO] --- maven-surefire-plugin:2.12.4:test (default-test) @ MavenAnsibleWebApp1 ---
[INFO] No tests to run.
[INFO]
[INFO] --- maven-war-plugin:3.4.0:war (default-war) @ MavenAnsibleWebApp1 ---
[INFO] Packaging webapp
[INFO] Assembling webapp [MavenAnsibleWebApp1] in [/home/sumanth/Devops/MavenAnsibleWebApp1/target/MavenAnsibleWebApp1]
[INFO] Processing war project
[INFO] Copying webapp resources [/home/sumanth/Devops/MavenAnsibleWebApp1/src/main/webapp]
[INFO] Building war: /home/sumanth/Devops/MavenAnsibleWebApp1/target/MavenAnsibleWebApp1.war
[INFO]
[INFO] --- maven-install-plugin:2.4:install (default-install) @ MavenAnsibleWebApp1 ---
[INFO] Installing /home/sumanth/Devops/MavenAnsibleWebApp1/target/MavenAnsibleWebApp1.war to /home/sumanth/.m2/repository/com/example/MavenAnsibleWebApp1/1.0-SNAPSHOT/MavenAnsibleWebApp1-1.0-SNAPSHOT.war
[INFO] Installing /home/sumanth/Devops/MavenAnsibleWebApp1/pom.xml to /home/sumanth/.m2/repository/com/example/MavenAnsibleWebApp1/1.0-SNAPSHOT/MavenAnsibleWebApp1-1.0-SNAPSHOT.pom
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 3.868 s
[INFO] Finished at: 2026-05-09T23:47:09+05:30
[INFO] -----
sumanth@sumanth-VirtualBox:~/Devops/MavenAnsibleWebApp1$
```

Step 6: Check the **tree** structure

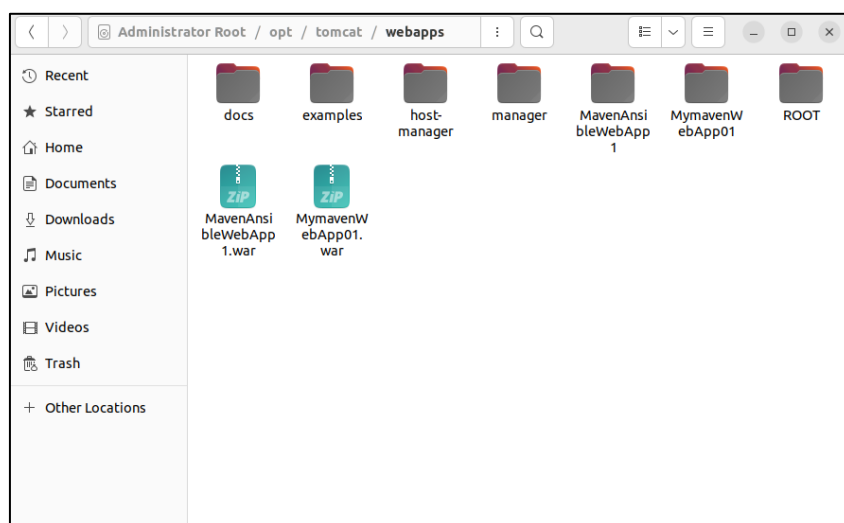
```
sumanth@sumanth-VirtualBox:~/Devops/MavenAnsibleWebApp1$ tree
.
├── ansible
│   ├── hosts.ini
│   └── playbook.yml
├── Jenkinsfile
├── pom.xml
├── src
│   └── main
│       ├── resources
│       └── webapp
│           ├── index.jsp
│           └── WEB-INF
│               └── web.xml
└── target
    ├── classes
    ├── MavenAnsibleWebApp1
    │   ├── index.jsp
    │   ├── META-INF
    │   ├── WEB-INF
    │   │   └── classes
    │   └── web.xml
    ├── MavenAnsibleWebApp1.war
    ├── maven-archiver
    └── pom.properties

13 directories, 10 files
```

Step 7: Run **sudo /opt/tomcat/bin/startup.sh** in order to start the Tomcat Server

```
sumanth@sumanth-VirtualBox:~/Devops$ cd ..
sumanth@sumanth-VirtualBox:~$ sudo /opt/tomcat/bin/startup.sh
[sudo] password for sumanth:
Using CATALINA_BASE:   /opt/tomcat
Using CATALINA_HOME:   /opt/tomcat
Using CATALINA_TMPDIR: /opt/tomcat/temp
Using JRE_HOME:        /usr
Using CLASSPATH:        /opt/tomcat/bin/bootstrap.jar:/opt/tomcat/bin/tomcat-juli.jar
Using CATALINA_OPTS:
Tomcat started.
sumanth@sumanth-VirtualBox:~$
```

Step 8: Now before running it, we need to copy the War file created and paste in the file path of tomcat file i.e. **/opt/tomcat/webapps**. After Pasting we can run the tomcat server easily



Step 9: Login in to the Respective Tomcat server with Username and Password for Manager App and click on **MavenAnsibleWebApp01**

/manager

localhost:9090/manager/html



Tomcat Web Application Manager

Message:

OK

Manager

List Applications

HTML Manager Help

Manager Help

Server Status

Applications

Path	Version	Display Name	Running	Sessions	Commands
/	None specified	Welcome to Tomcat	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/MavenAnsibleWebApp01	None specified	Archetype Created Web Application	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/MymavenWebApp01	None specified	Archetype Created Web Application	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/docs	None specified	Tomcat Documentation	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/examples	None specified	Servlet and JSP Examples	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes

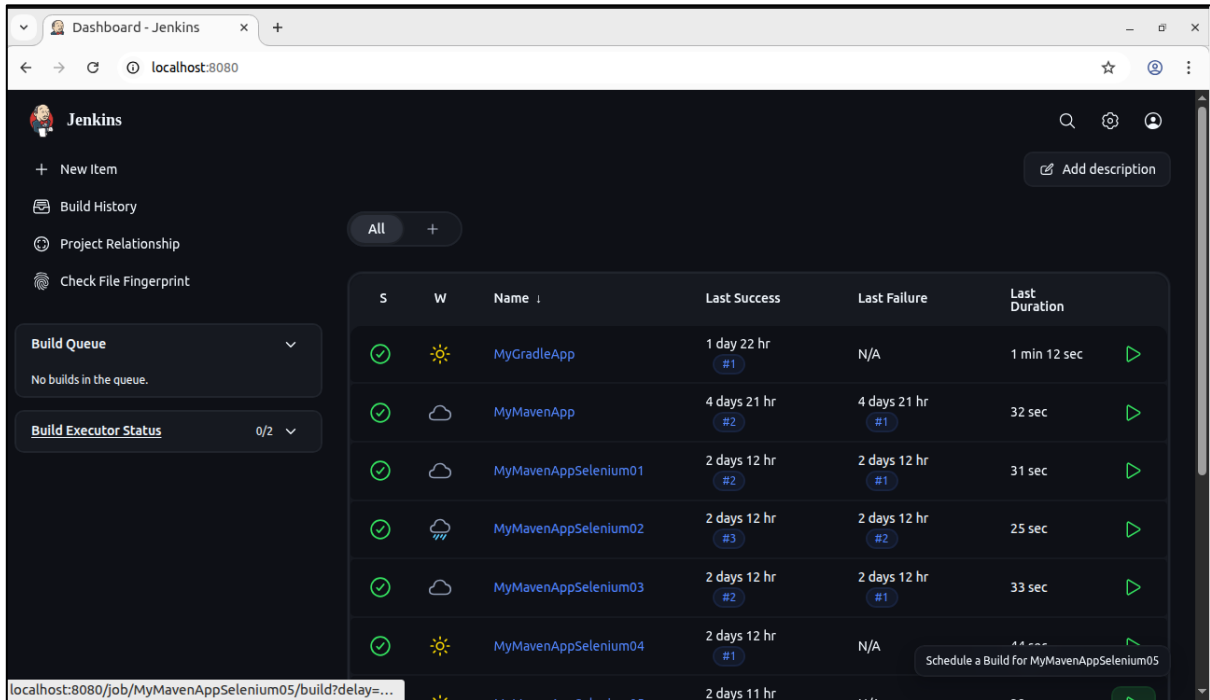
localhost:9090/Mymaver

localhost:9090/MymavenWebApp01/

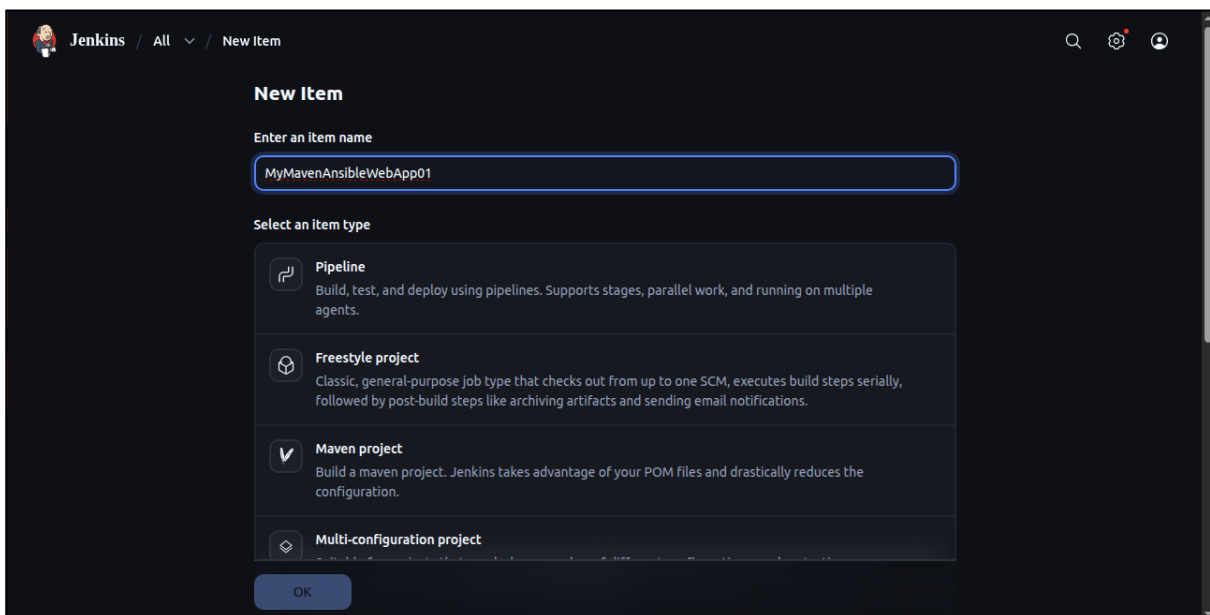
Sumanth R 1BI23CS218

Jenkins Pipeline for MyMavenAnsibleWebApp01

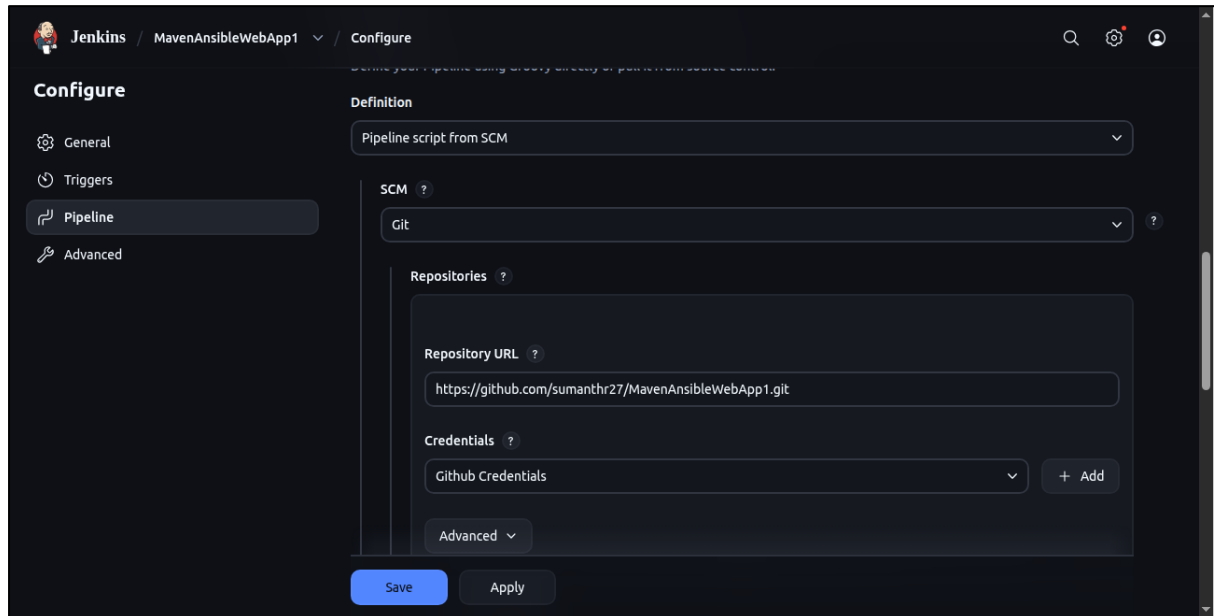
Step 1: Open the Jenkins Dashboard in order to create a new item



Step 2: Click on New Item in order for creating an New Item and thus give the name as “MyMavenAnsibleWebApp01”



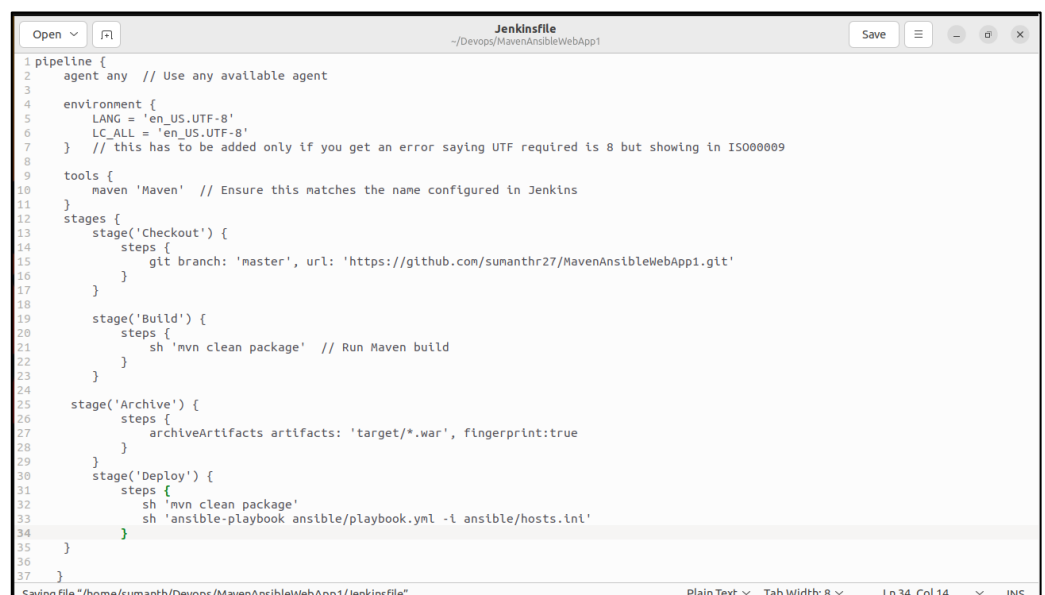
Step 3: Configure the settings by Defining the script as “Pipeline Script from SCM” and SCM as “Git”. Thus, now as we have selected Git, we need to paste the Repository URL of the project given in GitHub and choose Credentials as “GitHub Credentials” which was created earlier.



Step 4: Now in Terminal, go to directory created for MyMavenWebApp01 and create a new file called “Jenkinsfile”

```
sumanth@sumanth-VirtualBox:~$ cd Devops
sumanth@sumanth-VirtualBox:~/Devops$ cd MymavenWebApp01
sumanth@sumanth-VirtualBox:~/Devops/MymavenWebApp01$ ls
Jenkinsfile pom.xml src target
sumanth@sumanth-VirtualBox:~/Devops/MymavenWebApp01$ gedit Jenkinsfile
```

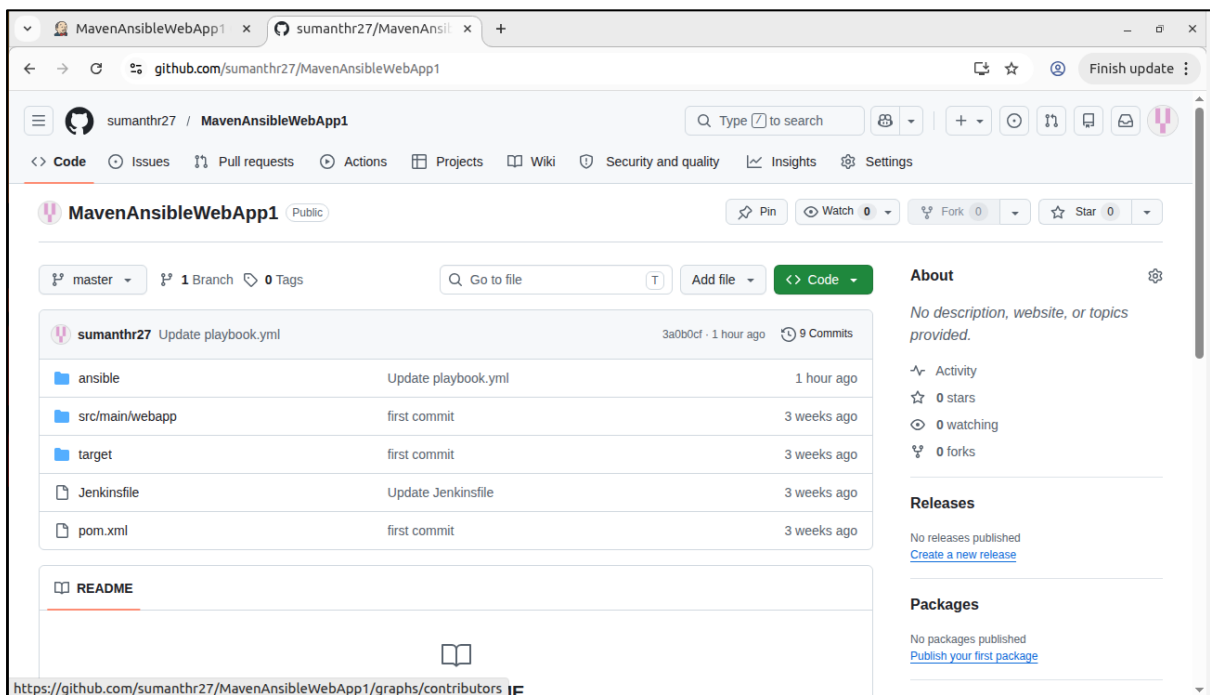
Step 5: Paste the available code inside the Jenkinsfile



Step 6: We need to update pom.xml also in order to change the versions of source and target from 1.7 to 1.8

```
<!-- Example: Maven Compiler Plugin to compile Java code -->
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-compiler-plugin</artifactId>
<version>3.8.1</version>
<configuration>
<source>1.8</source>
<target>1.8</target>
</configuration>
</plugin>
```

Step 7: Now we need to Push changes and commit by pushing the necessary “Jenkinsfile” to the respective Repo



The screenshot shows the GitHub interface for a repository named 'MavenAnsibleWebApp1' by user 'sumanthr27'. The repository is public and has 0 stars, 0 forks, and 0 watchers. The main content area displays a list of files and their commit history:

File	Commit	Time
ansible	Update playbook.yml	1 hour ago
src/main/webapp	first commit	3 weeks ago
target	first commit	3 weeks ago
Jenkinsfile	Update Jenkinsfile	3 weeks ago
pom.xml	first commit	3 weeks ago

The right sidebar contains sections for 'About' (No description, website, or topics provided), 'Releases' (No releases published), and 'Packages' (No packages published).

Step 8: Now go to Jenkins and Build the Pipeline and thus check the Stage view for the successful execution.

Jenkins

MavenAnsibleWebApp1

Search

Settings

Profile

Status

Changes

Build Now

Delete Pipeline

Configure

Rename

Full Stage View

Stages

Pipeline Syntax

MavenAnsibleWebApp1

Last Successful Artifacts

MavenAnsibleWebApp1.war

1.81 KiB

view

Stage View

Average stage times:
(full run time: ~1min 6s)

	Declarative: Checkout SCM	Declarative: Tool Install	Checkout	Build	Archive	Deploy
#8 May 09 22:24 2 commits	2s	504ms	2s	12s	973ms	17s
#7 May 09 22:10 1 commit	2s	769ms	2s	14s	1s	29s
#6 Apr 15 13:31 1 commit	2s	617ms	2s	23s	1s	22s failed
	2s	263ms	2s	16s	724ms	23s failed

Builds >

Filter

/

9 May 2026

✓

#8

22:24

✗

#7

22:10

45 April 2026